



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Developing AI Apps and Agents on Azure (AI-103)

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Conducted by

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Version 5.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.0	26 Aug 2026	Prior AI-102 courseware release	Tertiary Infotech Academy Pte Ltd
3.0	27 Sep 2026	AI-103 title transition	Tertiary Infotech Academy Pte Ltd
4.0	27 Sep 2026	Five-domain AI-103 courseware and original Tertiary labs	Tertiary Infotech Academy Pte Ltd
5.0	27 Sep 2026	Updated to v5.0 deck slide map, pinned Microsoft Learning lab sources, five AI-103 domains, and approved 60+60 minute assessments.	Tertiary Infotech Academy Pte Ltd

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Course Overview

A two-day instructor-led WSQ course that develops the ability to analyse, evaluate, and improve Azure AI solutions through Microsoft Foundry services, governed engineering decisions, and observable evidence.

Learning Outcomes

- LO1: Analyse algorithms in Microsoft Azure AI applications and establish their correlation with efficiency.
- LO2: Identify and evaluate strengths and limitations of Microsoft Azure AI applications.
- LO3: Assess feasibility and improvements when applying Azure AI to product and maintenance processes.

Daily Schedule

Time	Duration	Topic / Activity	Method	Slides
Day 1 · 09:00–09:40	40 min	Administration, outcomes, assessment briefing	Facilitated briefing	Slides 1–15
Day 1 · 09:40–10:45	55–70 min	Topic/Lab 01: Plan a Foundry project	Mechanism + evidence lab	Slides 17–34
Day 1 · 11:00–12:05	55–70 min	Topic/Lab 02: Evaluate models and release guardrails	Mechanism + evidence lab	Slides 35–53
Day 1 · 12:05–13:00	55–70 min	Topic/Lab 03: Build a generative chat app	Mechanism + evidence lab	Slides 54–71
Day 1 · 14:00–15:05	55–70 min	Topic/Lab 04: Build a Foundry agent	Mechanism + evidence lab	Slides 72–89
Day 1 · 15:05–16:10	55–70 min	Topic/Lab 05: Use a custom tool in an agent	Mechanism + evidence lab	Slides 90–107
Day 1 · 16:25–17:35	55–70 min	Topic/Lab 06: Develop a vision-enabled chat app	Mechanism + evidence lab	Slides 108–125
Day 1 · 10:45–11:00	15 min	Morning break	Break	—
Day 1 · 13:00–14:00	60 min	Lunch	Break	—
Day 1 · 16:10–16:25	15 min	Afternoon break	Break	—
Day 1 · 17:35–18:00	25 min	Evidence review and recap	Discussion	Slides 17–125
Day 2 · 09:00–09:15	15 min	Digital attendance and recap	Review	Slide 126
Day 2 · 09:15–10:20	65–75 min	Topic/Lab 07: Analyze text	Mechanism + evidence lab	Slides 127–144
Day 2 · 10:35–11:45	65–75 min	Topic/Lab 08: Develop a text analysis agent	Mechanism + evidence lab	Slides 145–162
Day 2 · 11:45–13:00	65–75 min	Topic/Lab 09: Extract multimodal information	Mechanism + evidence lab	Slides 163–181
Day 2 · 14:00–15:15	65–75 min	Topic/Lab 10: Build a knowledge mining solution	Mechanism + evidence lab	Slides 182–199
Day 2 · 10:20–10:35	15 min	Morning break	Break	—
Day 2 · 13:00–14:00	60 min	Lunch	Break	—
Day 2 · 15:15–15:45	30 min	Assessment briefing and preparation	Briefing	Slides 11–13
Day 2 · 16:00–18:00	120 min	WA-SAQ (60 min) + PP (60 min)	Open-book assessment	Slides 202–204

Topic-by-Topic Breakdown

Topic 01 - Plan a Foundry project · Slides 17–34

Lab 01 - Plan a Foundry project · Slide 25

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 02 - Evaluate models and release guardrails • Slides 35–53

Lab 02 - Evaluate models and release guardrails • Slide 44

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 03 - Build a generative chat app • Slides 54–71

Lab 03 - Build a generative chat app • Slide 62

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 04 - Build a Foundry agent • Slides 72–89

Lab 04 - Build a Foundry agent • Slide 80

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 05 - Use a custom tool in an agent • Slides 90–107

Lab 05 - Use a custom tool in an agent • Slide 98

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 06 - Develop a vision-enabled chat app • Slides 108–125

Lab 06 - Develop a vision-enabled chat app • Slide 116

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 07 - Analyze text • Slides 127–144

Lab 07 - Analyze text • Slide 135

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 08 - Develop a text analysis agent • Slides 145–162

Lab 08 - Develop a text analysis agent • Slide 153

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 09 - Extract multimodal information • Slides 163–181

Lab 09 - Extract multimodal information • Slide 172

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 10 - Build a knowledge mining solution • Slides 182–199

Lab 10 - Build a knowledge mining solution • Slide 190

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Resources Required

- Trainer PPTX and learner PDF
- Learner Guide
- Azure subscription or trainer-provided lab environment
- Modern browser, Azure CLI, Python 3, and Visual Studio Code as required
- LMS/TMS access for courseware and assessment

Assessment

Approved instruments: Written Assessment - Short-Answer Questions (WA-SAQ), 60 minutes, K1–K6; Practical Performance (PP), 60 minutes, A1–A6. Both are individual and open book. Complete assessment digital attendance before starting and submit candidate papers on the LMS.